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Disaster management and its importance: a review paper of Dakshina Kannada and Kodagu districts, Karnataka, India

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Abstract. An extreme occurrence within the earth's system that causes human fatalities, injuries, and property destruction is referred to as a "disaster" from a social perspective. Disaster management is one of the most critical topics around the world. Many countries are getting affected because of its high magnitude impact on economy and loss of life. The Karnataka government must always be on guard and ready to meet problems due to the frequent natural disasters like landslides, floods, droughts, and human-induced dangers. In order to lessen the effect of disasters on the community, the Government of Karnataka has built enhanced institutional structures for disaster risk reduction. The main aim of the review paper of this, is to create awareness about the situations occurring due to any form of disasters and to develop preparedness in younger generation to face disaster situations.

1. Introduction

According to science, at least one or more geological phenomena or earth system processes are actively, continuously, and cyclically involved in every natural disaster that has ever occurred on the earth surface of the world. These natural occurrences or processes are referred to as "Geohazards" because they pose a risk to people, other living things, their property, and the environment (Rashmi Jain et al, 2019). Even though disaster is a common term yet people are not aware how to be prepared and safe. On the other hand, even though the NGOs and government are providing facilities. As a citizen of India, it is our duty to take initiative for helping our society (Raj Nirmal et al, 2020).

The ideas of Disaster Risk Management (DRM) must be incorporated into every sector of growth. This would guarantee that all construction projects - roads, airports, canals, hospitals, schools, bridges, and railways—are constructed in accordance with the necessary standards and enhance the resilience of the communities they aspire to serve. This underlines the requirement to ensure that all infrastructure development complies with the highest requirements for disaster safety. Such a tactic is wise and will be profitable in the long run. It is essential that disaster risk factors be taken into account in all public investments. Such prospects are represented in India by the "housing for everyone" programme and "smart cities" efforts. The Karnataka State Disaster Management Plan (KSDMP) has been created and updated annually with the assistance of the State Executive

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Committee in order to streamline and direct the state's disaster management actions (SEC). The KSDMP has been supplying pertinent information to the parties involved in disaster response (KSDMP 2020-21). Along with KSDMP, each district's DDMP preparation is a "dynamic document" that it will continually be revised in accordance with district best practices and knowledge base (DDMP 2022-23).

1.1 Paradigm shift

A paradigm change in disaster management in India has been institutionalised through the DM Act of 2005 and the NPDM of 2009. This represents a transition from a focus on emergency reaction to one on proactive prevention, mitigation, and preparedness. While natural disasters cannot be prevented, the NPDM emphasises that they may be averted from becoming catastrophes with the right mitigation and risk reduction methods. Hazards combine with social, physical, economic, and environmental vulnerabilities to create disaster risk. The NPDM recommends a multifaceted strategy including the following for disaster risk reduction and mitigation:

- Implementing risk mitigation strategies across all development initiatives.
- Initiating mitigation initiatives via cooperative efforts of the Central & State Governments in designated high precedence regions.
- Promoting and helping mitigation efforts at the State level.
- Examining natives' understanding of disasters and coping techniques.
- Giving heritage structure protection the weight, it deserves.

1.2 Natural Hazards

The Disaster Information Management Methodology of DesInventar uses a "universally recognised categorization system that divides catastrophes occurring from natural hazards into five primary groups and is utilised internationally for the Sendai targets monitoring"

1. **Geophysical:** Geological processes or phenomena that may result in property destruction, loss of livelihoods or other health effects, disturbance of social or economic order, or environmental harm. Some of these processes have significant contributions from hydro-meteorological aspects.
2. **Hydrological:** Events brought on by variations in the regular water cycle and/or overflowing of water bodies as a result of favorable wind conditions.
3. **Meteorological:** Short-lived/small to meso-scale atmospheric events that occur (in the spectrum from minutes to days).
4. **Climatological:** long-lived meso- to macro-scale processes that result in certain events (in the spectrum from intra-seasonal to multi-decadal climatic variability).
5. **Biological:** Exposure to pathogenic microorganisms, bioactive, and toxic substances that may result in loss of life, illness, injury, or other health effects, property damage, loss of livelihoods and services, social and economic disruption, or environmental damage is an example of an organic process or phenomenon that is spread by biological vectors.

2. Kodagu at a glance

One among the smallest district in Karnataka is Kodagu which is also known as Coorg. It is the most beautiful hill station in south-west Karnataka region. This place is also called the coffee cup of India. The main source of income for this place is agriculture and tourism. The town capital of Kodagu district is Madikeri with an estimated population of 30000.

Torrential rains and floods accompanied by landslides in Kodagu district have brought agriculture to a standstill. The district's economy is based on agriculture, which produces rice, pepper, cardamom, and coffee. In 2017–18, 45,000 growers produced 1,16 lakh tonnes of Arabica and Robusta coffee. Kodagu contributes significantly to the production of fragrant soft drinks and provides 40% of the country's coffee. Farmers in Kodagu anticipated a big crop in 2018–19, but excessive rain and flooding damaged over 60% of the district's berries, and topsoil washed away in the majority of places.

and no longer be able to discern the farmed terrain. More than 150 landslide-prone locations have been identified by the Geological Survey of India in Kodagu alone.

Due to its proximity to the shore, Kodagu Prefecture has a typical tropical climate with mild to moderate humidity. High humidity, frequent downpours, and a chilly summer are some of its characteristics, making it recognised to be fairly pleasant and healthful. A monsoon season that lasts from early June to late September occupies a large portion of the year. Even "certain portions of the district receive a large quantity of rain" in October and November after the monsoon. According to an analysis of data from the previous ten years, the district's westernmost and most heavily wooded Madikeri taluk has the most rainfall (an average of 3302.46 mm), and its easternmost and driest portions (average 2105.22 mm). The taluk in Somvarpet is the region of the district that is farther north and has less forest cover; it is also close to the Plains.

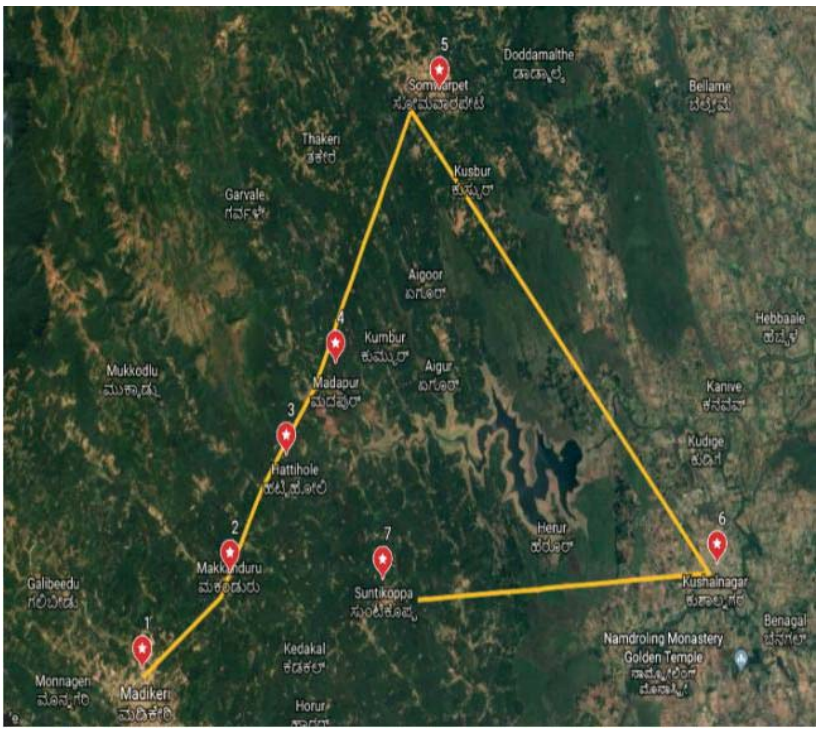

Figure 1. Disaster prone locations in Kodagu District

3. Dakshina Kannada at a glance

The headquarters of Dakshina Kannada are located at Mangaluru. Mangaluru Taluk is traversed by the Netravati River and the Gurupura River, the two principal rivers in the region. The district and Mangaluru in particular are well connected by road, rail, sea, and air. The Konkan Railway and National Highway 66 (Kanyakumari to Mumbai) both travels north to south along the district's coastline. From west to east, NH 75 traverses the district as it travels from Mangaluru to Bangalore.

Landslide occurrence is a frequent and recurring phenomenon in Western Ghats especially during monsoon. A number of landslides occurred in the isolated mounts and hill ranges of Western Ghats during 2018 and 2019 even at present, resulting in loss of lives and extensive damage to roads, dwelling units and other properties even death, uprooting of lot of trees and disrupting traffic movements. The disastrous event created panic amongst the general public and generated serious concern among the administrators, scientists and engineers. District administration carried out rapid assessment of the landslide event as well as the previous events along the three important road

corridors of Dakshina Kannada and the adjoining districts, Karnataka, from during January 2019. The various studies taken were to find out type, causative factors, magnitude of the landslides for arriving control and corrective measures and also to identify critical landslides and slopes that warrants detail study and workout long term to control or preventive measures.

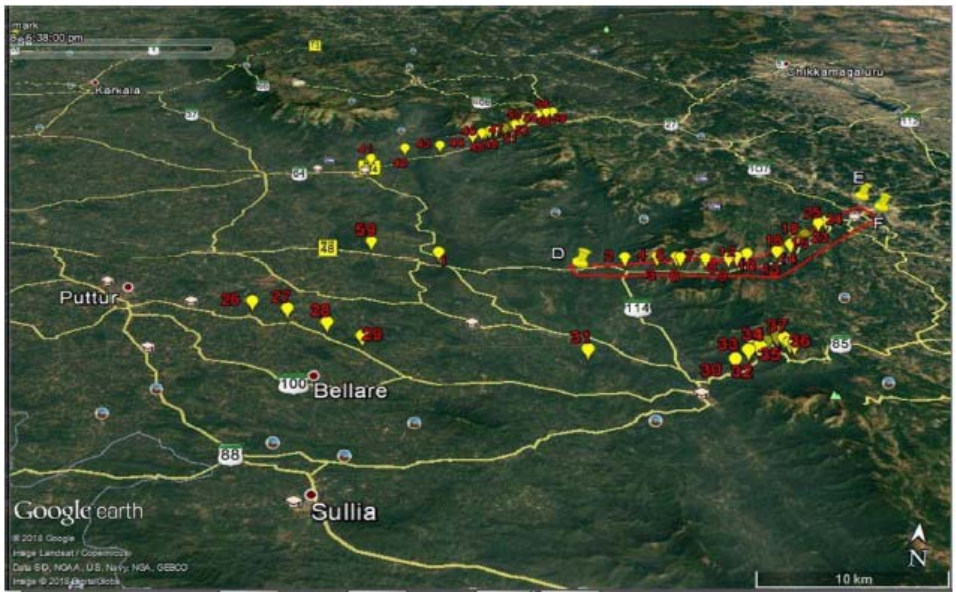

Figure 2. Map of slide/slide prone locations along NH 48 and NH 234 and SH 85

4. Priorities; Targets

India is a member to the Sendai Framework, a voluntary, non-binding 15-year agreement that acknowledges the state's important role in disaster risk reduction but also that, the duty must be shared with other stakeholders, such as local Government, the corporate sectors, and others.

The following four areas of action are prioritised under the Sendai framework:

1. Being aware of catastrophe risk.
2. Increasing the effectiveness of catastrophe risk management.
3. Making investments to reduce catastrophe risk to increase resilience.
4. Improving catastrophe readiness to provide an efficient response and "Build Back Better" in terms of recovery, rehabilitation, and reconstruction.

Figure 3. Sendai Framework for Disaster Risk Reduction (DRR) - 7 Global Targets

5. Conclusion

- Public have lack of knowledge about how to deal with a disaster.
- Children should be taught about coping with a disaster during the primary education itself.
- Disaster preparedness should be added to the curriculum.
- Disaster prone areas need special drainage systems.
- Disaster preparedness should be done as an activity in rural areas.

Despite being a cross-cutting characteristic, the accountability framework now includes a specific thematic action area for social inclusion, along with a list of related thematic sub-areas of action.

The catastrophe serves as a serious wake-up call for the local populace and the government to protect the ecology and stop the overuse of natural resources. The whole Western Ghats have been divided into three separate categories of ecologically sensitive zones, each of which has varied degrees of restrictions on development, according to the 'Gadgill' commission report. By protecting natural resources from overexploitation, these proposals might help the Western Ghats' ecological deterioration. The Western Ghats ecology continued to deteriorate as a result of the government's "negligence to issue" these regulations, which made it easier for similar calamities to happen.

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